DHRUV MEHTA

Civil Engineering Student · Structures

Fredericton, NB · mehtadhruv872@gmail.com · linkedin.com/in/mehtadhruv872

PROFILE

Fourth-year civil engineering student at the University of New Brunswick (BSc 2027) focused on structures. OSIM-certified bridge inspector with hands-on experience in structural steel project management, structural calculations, and bridge health monitoring research. Comfortable on site, in the design office and in the lab, and used to leading student teams.

PROFESSIONAL EXPERIENCE

Undergraduate Research Assistant · Structural Health Monitoring Lab, University of Waterloo — Summer 2026

- Selected for the Rising Stars Program, Faculty of Engineering; supervised by Dr. Alana Lund
- Indirect structural health monitoring of bridges using a mobile robotic sensing platform (Husky robot)
- Designed the robot characterization protocol and led field data collection on two pedestrian bridges
- Performed modal analysis and model updating; wrote SOPs and test procedures; maintained ROS 2 control code and Python analysis

Engineering Student / Bridge Inspector · Government of New Brunswick, DTI/MTI, Fredericton — 2024 – 2025

- Inspected bridges and related structures for condition and safety using OSIM
- Tested structural components including pilings and abutments; documented observations and test results
- Wrote inspection reports with recommended maintenance, repairs and budget allowances; maintained inspection records

Project Manager · Ocean Steel & Construction Ltd., Saint John — 2022 – 2023

- Scoped structural steel projects from contract drawings and project requirements
- Scheduled fabrication and coordinated foremen, crews, trades, suppliers and general contractors
- Tracked RFIs, change orders and POs; reported monthly and quarterly project and financial performance to senior management

Structural Associate · Varcon Inc., Fredericton — 2021 – 2022

- Completed engineering calculations for weights and loads, by hand and in software
- Reviewed drawings from CAD technicians and verified construction documents against client requirements

PROJECTS & COMPETITIONS

Howe Truss Bridge · CE 3983 Civil Design II, team of four — Apr 2026

- Chose a Howe truss over Warren and Parker options; designed members to CSA O86-24 and analyzed member forces
- Built and load-tested to failure at 5.22 kN; 1st in embodied carbon (481 g CO_2), 3rd in strength-to-cost ($74.73)

Stuttgart 21 as a Model for Moncton · CE 5242 Transportation Field School, Germany — 2026

- Solo technical report on transit-led land value capture; proposed a BRT corridor on Vaughan Harvey Boulevard to the Vision Lands, funded by municipal land sales in place of TIF

Troitsky Bridge Building Competition · Concordia University, Montréal — 2025, 2026

- Team captain of UNB's Team UNBreakable both years: design schedule, build planning and competition day

TimberFever Design-Build · Full-scale timber pavilion, team of six — 2026

Global Sustainability Project Bootcamp · How to Change the World — Mar 2025

- Best Overall (Paris) with Team 2-M; the only engineer on a five-university team; designed a textile-reuse hub concept for Paris

LEADERSHIP

VP Competitions (2025, 2026) · VP External (2024) · CSCE Student Chapter, UNB 2024 – now

- Help run the chapter's events, industry nights and competition teams; building UNB's 2027 Troitsky team

EDUCATION

Bachelor of Science in Civil Engineering · University of New Brunswick 2023 – 2027

- Certificate of Academic Excellence, highest cumulative GPA (2025–26) · Nels Anderson II Scholarship (2025–26)
- Edwin Jacob Special University Scholarship (2024–25) · Allan K. Grimmer Prize (2024–25) · Dean's List (2023–24, 2024–25)

Diploma, Building Engineering Technology: Structural · New Brunswick Community College 2019 – 2021

- Graduated with Honours

CERTIFICATIONS & SKILLS

Certifications: OSIM Bridge Inspector (Stantec, BMS) · WHMIS 2015 · Fall Arrest & Rescue · OH&S Due Diligence (NB)
Design & analysis: STAAD.Pro 2025 · S-Frame · AutoCAD 2D/3D · Revit · CSA O86
Construction management: Procore · Bluebeam Revu · RFIs & change orders · Contract management · Estimating
Research: ROS 2 · Python · Modal analysis · Field testing
Languages: English · French · Hindi · Gujarati

References available on request.